WORKED SOLUTION

STEP III 1990, Question 13

Mechanics • Projectile motion

STEP III 1990, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q13

The key idea

There are two stages: circular motion on the sphere until the normal reaction vanishes, then free flight as a projectile. The condition for leaving fixes the speed at the moment of leaving in terms of θ, and then the hint does the rest: write down the condition for the projectile to be back on the sphere, $x^2 + y^2 = a^2$, as an equation in t. It collapses beautifully, because the leaving condition makes several terms cancel.

Solution

The model

We treat P as a particle and the sphere (radius a) as smooth, so the only forces on P while it is in contact are its weight mg and the normal reaction R, directed towards O. The motion takes place in a vertical plane through O; we take axes with origin O, x horizontal and y vertically upwards, and let P climb the right-hand side of the sphere.

For P to leave the surface at all, it must do so above the level of O: below that level the component of gravity along PO points away from O, so R cannot be zero there while P moves on the circle. Hence $0 < \theta < \frac{\pi}{2}$.

Leaving the surface

Let P have speed v when OP makes angle θ with the upward vertical, so P is at height $a\cos\theta$ above O. Resolving towards O, the weight has component $mg\cos\theta$ in that direction, so

$$R + mg\cos\theta = \frac{mv^2}{a}.$$

P leaves when $R = 0$, so at that moment

$$v^2 = ag\cos\theta.$$

(Conservation of energy shows that the speed of projection must be u with $u^2 = v^2 + 2ga(1+\cos\theta) = ag(2+3\cos\theta)$, so every θ in $(0, \frac{\pi}{2})$ is achievable by choosing u suitably. As P climbs, v falls and R falls, so R first vanishes exactly at this point.)

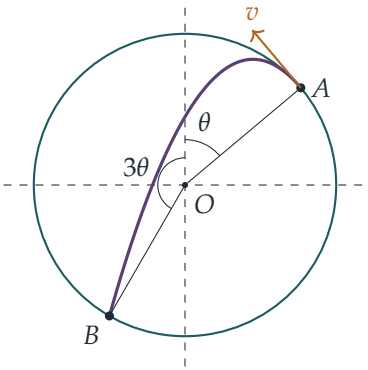
The free flight

At the moment of leaving, call it $t = 0$, P is at $(a\sin\theta, a\cos\theta)$, and its velocity is along the tangent, perpendicular to OP and pointing up and over the top:

$$\text{velocity} = v(-\cos\theta,\ \sin\theta).$$

After that only gravity acts, so

$$x = a\sin\theta - vt\cos\theta, \qquad y = a\cos\theta + vt\sin\theta - \tfrac{1}{2}gt^2.$$



P leaves at A and, as we now show, strikes the sphere again at B.

P is on the sphere when $x^2 + y^2 = a^2$. Expanding, the terms that are constant or linear in t are

$$a^2\sin^2\theta + a^2\cos^2\theta = a^2 \qquad \text{and} \qquad -2avt\sin\theta\cos\theta + 2avt\cos\theta\sin\theta = 0,$$

so

$$\begin{aligned} x^2 + y^2 - a^2 &= v^2t^2\cos^2\theta + v^2t^2\sin^2\theta - gt^2\left(a\cos\theta + vt\sin\theta\right) + \tfrac{1}{4}g^2t^4 \\ &= \left(v^2 - ag\cos\theta\right)t^2 - gv\sin\theta\, t^3 + \tfrac{1}{4}g^2t^4 \\ &= gt^3\left(\tfrac{1}{4}gt - v\sin\theta\right), \end{aligned}$$

using $v^2 = ag\cos\theta$ in the last step.

Ask yourself Why does $t = 0$ come out as a triple root, and not just a single one?

At the moment of leaving, gravity alone is providing exactly the acceleration v^2/a towards O needed to move on the circle. So the parabola starts off with the same

direction *and* the same curvature as the sphere: it hugs the surface very closely before peeling away inside it. That is why the t^2 term cancels as well as the t term.

For $0 < t < 4v\sin\theta/g$ the expression is negative, so P is strictly inside the sphere, and it next meets the surface at

$$T = \frac{4v\sin\theta}{g}.$$

Where it lands

At $t = T$, using $vT\sin\theta = 4v^2\sin^2\theta/g = 4a\cos\theta\sin^2\theta$ and $\frac{1}{2}gT^2 = 8v^2\sin^2\theta/g = 8a\cos\theta\sin^2\theta$,

$$y = a\cos\theta + 4a\cos\theta\sin^2\theta - 8a\cos\theta\sin^2\theta = a\cos\theta\left(1 - 4\sin^2\theta\right).$$

(This is $a\cos 3\theta$, since $\cos 3\theta = 4\cos^3\theta - 3\cos\theta = \cos\theta(1 - 4\sin^2\theta)$; similarly $x = -a\sin 3\theta$, so the impact point B is at angle 3θ from the upward vertical on the far side, as in the diagram.)

P strikes below the level of O exactly when $y < 0$. Since $0 < \theta < \frac{\pi}{2}$, $\cos\theta > 0$, so

$$y < 0 \iff 1 - 4\sin^2\theta < 0 \iff \sin\theta > \tfrac{1}{2} \iff \theta > \tfrac{\pi}{6},$$

the last step because sin is increasing on $(0, \frac{\pi}{2})$. When $\theta = \frac{\pi}{6}$ exactly, P strikes the sphere exactly level with O. So the angle that θ must exceed is

$$\boxed{\theta = \frac{\pi}{6} = 30^\circ}$$

and every θ with $30^\circ < \theta < 90^\circ$ does give an impact below the level of O.

Checks

- $\theta = 60^\circ$: the formula gives $y = a\cos 180^\circ = -a$, so P lands back at the lowest point, the point of projection. With $\theta = 45^\circ$ it lands at 135° from the upward vertical; both are $y < 0$, as they should be.
- As $\theta \to 0$, $T \to 0$ and $B \to A$: a particle that only just leaves at the top falls straight back onto the surface.
- A numerical simulation (stepping the motion on the sphere until $R = 0$, then the projectile until it meets the sphere) for $\theta = 10^\circ, 25^\circ, 29^\circ, 30.5^\circ, 31^\circ, 45^\circ, 60^\circ, 80^\circ$ matched $T = 4v\sin\theta/g$ and $y = a\cos 3\theta$ to four decimal places; for example $\theta = 31^\circ$ gives $y/a = -0.0523$.

Common slips

- Getting the direction of the velocity wrong: it is perpendicular to OP and still heading up and over, with components $(-v\cos\theta,\ v\sin\theta)$, not $(v\sin\theta,\ v\cos\theta)$.

- Forgetting to use $v^2 = ag\cos\theta$ before solving $x^2 + y^2 = a^2$, which leaves an awkward quartic instead of a clean factorisation.
- Dividing $1 - 4\sin^2\theta < 0$ through without noting that $\cos\theta > 0$ because P leaves above the level of O.

Beyond STEP

The triple root at $t = 0$ is a statement about *osculation*: the parabola and the circle share their tangent and their curvature at the leaving point, so they have contact of second order. Curvature and the osculating circle are central ideas in the differential geometry of curves, and they reappear in mechanics whenever we split acceleration into tangential and normal parts. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats motion in polar coordinates and constrained motion of this kind carefully.

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